ABHIJITH R PILLAI

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Summary

A Bachelor of Engineering in Artificial Intelligence and Machine Learning student at T. John Institute of Technology (2023–2027) with a solid foundation in IoT systems, AI-driven product design, and assistive technology. Skilled in Python, cloud computing, and embedded systems, with hands-on experience developing socially impactful innovations recognized at international hackathons.

Education

T. John Institute of Technology Bengaluru, Karnataka 2023-2027
Bachelor of Engineering in Artificial Intelligence and Machine Learning
Coursework: Data Structures & Algorithms, Discrete Mathematics, Python for Data Science, Azure Fundamentals, Operating Systems, Computer Organization & Digital Design.
Expected CGPA: 8.0

Projects

SmartBurn Crawler – Integrated Agricultural Residue Management System *(Finalist, AgriSpark International Hackathon – Bangkok, 2025)*

- Contributed ~55% to overall system design, focusing on functional integration of cutting, shredding, combustion, and emission control stages.
- Played a key role in crawler-based mobility and power integration design, improving operational stability on sloped terrains (20°–30°) by an estimated ~40%.
- Supported safety-oriented design decisions that reduced operational risk in uneven agricultural fields by ~35% (conceptual impact estimation).
- Selected as a Top 10 Finalist among 200+ international teams, highlighting strong real-world impact and innovation in sustainable agriculture.

Ripple Health - AI-Based Smart Wearable System (Prototyping & Ongoing Research Project)

- Led ~70% of the end-to-end system design, including sensor selection, wearable architecture, and data flow planning.
- Improved proposed health monitoring efficiency by ~40% through optimized sensor integration and reduced redundant data sampling.
- Project aims to create measurable impact in personal safety and preventive healthcare through scalable wearable technology.

AI-Based Exam Paper Corrector (Prototyping & Research Work)

- Completed core research work including problem identification, literature study, and AI-based evaluation methodology design.
- Contributed ~65% to research formulation, dataset planning, and NLP-driven answer similarity framework.
- Research outcomes indicate a potential ~60% reduction in manual correction time and ~45% improvement in evaluation consistency.

Social Media Accident Severity Analysis

- Led ~60% of system development for an AI-based extension that analyzes real-time social media posts to detect road accidents and related hazards.
- Implemented sentiment analysis, keyword intensity scoring, and geo-context filtering to assess accident severity and urgency.
- Achieved an estimated ~75% accuracy in classifying high-severity vs low-severity incidents during validation.

Speech Therapy Platform – AI-Based Voice Rehabilitation Assistant

- Contributed ~60% to system architecture and AI model integration for speech recognition and pronunciation analysis.
- Implemented speech analysis models achieving ~92% accuracy in word recognition and tone detection.
- Enabled measurable improvement in user fluency and pronunciation consistency through real-time feedback mechanisms.

Aquaponic Tank with Automatic Fish Food Dispenser *(Finalist)*

- Led ~70% of system design and IoT integration for a sustainable aquaponic ecosystem combining hydroponics and automated feeding.
- Integrated water-quality sensors achieving real-time monitoring accuracy of ±1°C (temperature) and ±0.1 pH.
- Recognized as a finalist for innovation, sustainability, and scalability in smart agriculture solutions.

SmartFarm AI - Intelligent Farming Decision Support System

- Developed an AI-driven farming platform enabling smarter decisions from cultivation to market
- Combined disease detection, irrigation insights, and crop guidance using real-time data
- Designed a scalable system supporting low-resource farmers and future drone-based monitoring

Skills

- **Technical Skills**
  Languages: Python, C, JavaScript, SQL
  AI/ML & Data Science: Machine Learning, Data Analysis, Computer Vision, NLP, Statistical Analysis, Sentiment Analysis
  Frameworks & Libraries: TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy
  Web Development: HTML, React.js, Node.js, PHP
  Databases: MySQL, MongoDB
  Tools & Platforms: Git, GitHub, Jupyter Notebook, Power BI, Tableau, Excel
  Core Concepts: API Integration, Software Development Life Cycle (SDLC), Debugging
- **Professional Skills**
  Problem Solving, Analytical Thinking, Data-Driven Decision Making, Collaboration, Agile Execution

Activities & Achievements

- Finalist, AgriSpark International Hackathon (Bangkok, 2025) - Represented India; SmartBurn Crawler ranked among top global teams
- 1st Place, TJ Hack IT - Social Media Accident Severity Analysis
- Finalist, Trikaya Hack to Hire (2025) - Ripple Health selected among 4000+ teams
- Finalist, Inception Prototype Expo (2024) - IoT-based Aquaponic System
- Participated & Volunteered, ICICNIS 2025 - AI/ML research conference
- Event Volunteer, QCFI 2024 - Cybersecurity and quality control sessions

Certification

- Microsoft Learn - AI, Azure & Cloud Fundamentals (30+ modules)
  Covered Azure AI Services, Computer Vision, NLP, Generative AI, Cloud Architecture, Security, and Data
- Advanced Python - Infosys Springboard
- Junior Software Developer - PMKVY
- Machine Learning - Innomatics Research Lab